

Gonalo Calvo

Software Engineering Student

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Skills

- **Programming:** C++, C#, Python, TypeScript, JavaScript, Java, Laravel
 - **Frontend:** Vite, CSS Modules, GSAP, Web Animations API, Accessibility (ARIA, WCAG)
 - **Backend:** FastAPI, SQLAlchemy (async), Alembic, Celery, Redis, JWT Auth
 - **AI/ML:** LLM integration (Gemini 2.0 Flash), vector embeddings, ChromaDB (semantic search)
 - **Tools:** Git, Docker, Docker Compose, Linux, SQL, PostgreSQL
 - **Concepts:** OOP, Algorithms and Data Structures, REST APIs, Async Pipelines, Component Architecture
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Projects

Dual-Design Portfolio Website

TypeScript, Vite, GSAP, CSS Modules, Howler.js

Built a production-spec portfolio platform with two fully distinct views — a Professional Executive View and an interactive PS Vita-inspired immersive view — switchable via a persistent toggle. Engineered a custom state management system using localStorage and custom DOM events, implemented FOUC prevention, and optimised for Core Web Vitals (LCP < 1.2s, CLS 0.00). Both views share a single JSON data source with strict TypeScript schemas.

Automated Content Continuity Pipeline (ACCP)

Python, FastAPI, PostgreSQL, ChromaDB, Gemini AI, Docker

Designed and built a production-grade async backend that uses LLM-powered semantic analysis to detect continuity conflicts across versioned documents. Architected a decoupled pipeline with FastAPI, Celery/Redis for task processing, and ChromaDB for vector search. Implemented JWT auth, SHA-256 document deduplication, multi-format text extraction (PDF, DOCX, MD), and full schema migration with Alembic.

Personal Projects (Unity/C#)

Developed a pathfinding system using the A* algorithm for dynamic obstacle navigation, and a procedural level generation system using modular room adjacency rules. Gained hands-on experience with real-time algorithmic problem solving and object-oriented design in C#.

Education

Bachelor in Informatics and Computing Engineering Faculdade de Engenharia da
Universidade do Porto (Graduation date: 2026)

Languages

- English — Fluent (C2)
- Portuguese — Native